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## “AI in educational decision making and data analytics”

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## Abstract

The present report focuses on the disruptive nature of the Artificial Intelligence (AI) in educational decision-making and data analytics within K-12 and higher education institutions. This paper provides a systematic review of seven research articles and investigates how AI-based tools can improve student performance forecasting, personalized learning, and learning analytics. The most interesting conclusions are the following: AI technologies, especially machine learning algorithms and predictive models can greatly enhance educational results in case of the adequate implementation. Nevertheless, there are still obstacles such as the shortage of infrastructure, educator training needs, ethical implications regarding the privacy of data and algorithm-based bias. The review notes that effective AI implementation involves moderate measures that will involve both the development of technologies and pedagogical skills. The report is relevant to the discussion of the best approaches to the implementation of AI in education as well as the key issues of equity, accessibility, and responsible use of technology. It gives recommendations on where future research should be conducted and how it should be implemented.

**Keywords:** Artificial Intelligence, Educational Decision-Making, Learning Analytics, Predictive Modeling, Educational Data Mining, K-12 Education

## 1. Introduction

The field of AI is changing the way educators use student data to make decisions in education and how to maximize learning outcomes. Predictive analytics with at-risk students, smart tutoring systems customizing teaching, and much more: AI technologies are transforming the educational landscape on an unprecedented scale. This systematic review is based on seven pioneered studies exploring a multifaceted effect of AI on educational decision-making in K-12 and higher education settings on an international scale.

Although AI shows great possibilities. (He, et al., 2025)obtained a predictive precision of 84% in optimizing the teaching of mathematics, and reported substantial improvements in performance in STEM disciplines, implementation of critical impediments. The lack of infrastructure, poor teacher preparation, and data quality issues along with ethical issues related to privacy and algorithmic bias are critical. The article by (Wang, 2021)is characterized as a warning, as the author highlights the need to consider the efficiency benefits of AI and balance them with the potential of increased biases, eroded moral judgment, and security threats. The report contains evidence-based information on the existing AI applications, machine learning techniques, performance indicators, commercial solutions, and implementation issues, as well as presents feasible solutions to the responsible AI integration, combining technological advancement with pedagogical skills and educational equality.

## 2. Aims and Objectives

### Aims

In order to critically analyze the role and influence of AI technologies in teaching and learning decisions and data analytics, determine best practice, challenges, and paths forward to implement AI technologies in education.

### Objectives

1. To examine different AI-based methods that have been applied in the educational domain to make decisions and analytics.
2. To assess the efficacy of the AI applications in enhancing learning and student performance.

3. To determine major difficulties and constraints in the application of AI technologies in educational organizations.
4. To analyze the ethical issues of using AI in education such as privacy, bias, and fairness.
5. To give evidence-based suggestions that can be used in the integration of AI in the education process of making decisions.

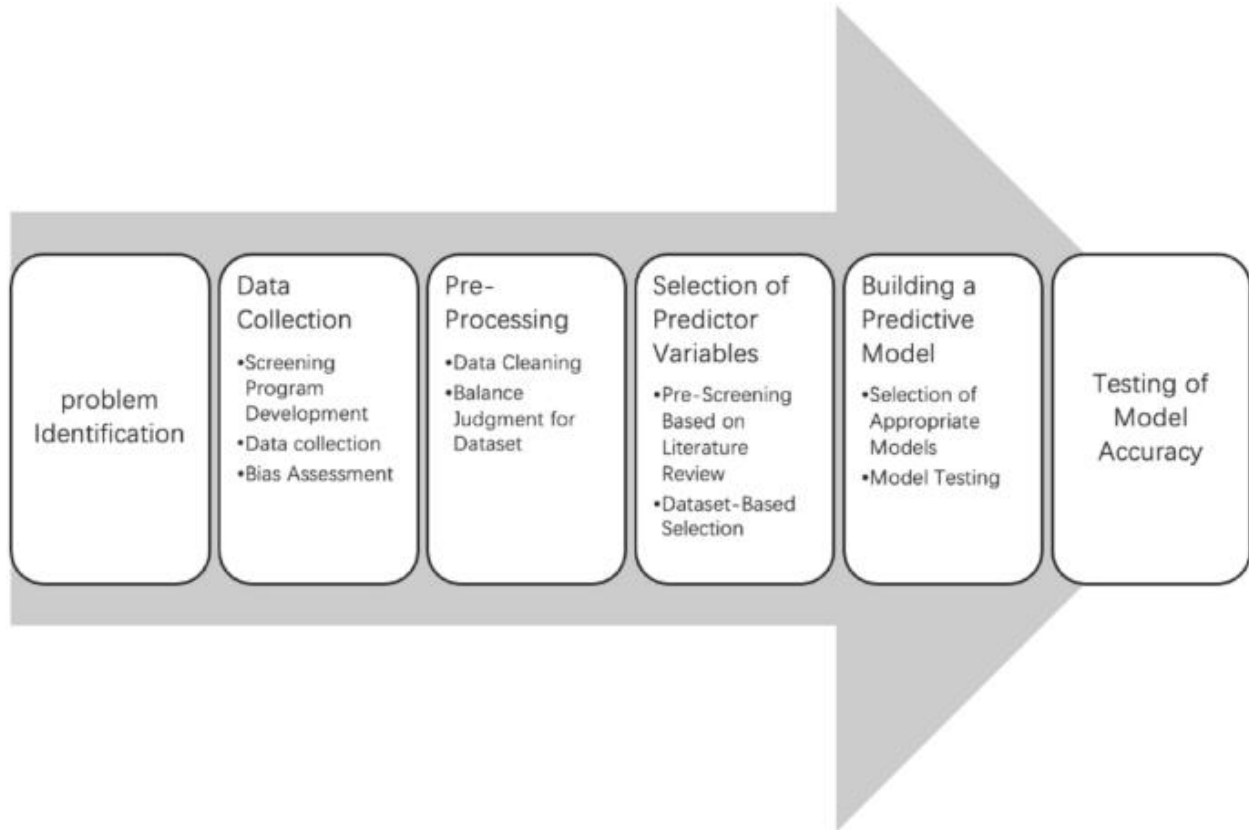
### **3. AI Training Techniques**

#### **3.1 Machine Learning Foundations**

AI as applied in educational decision-making is largely based on machine learning algorithms that can process large amounts of educational data to produce patterns and predictions. According to Ahmad et al. (2024), four main categories of machine learning that are used in the educational setting include supervised learning, unsupervised learning, reinforcement learning, and generative AI. Educational applications are dominated by supervised learning, which is used in most applications in performance prediction and dropout forecasting because it is effective in the case of labeled data. Salas-Pilco et al. (2022) showed that the algorithms including the Random Forest, Support Vector Machines, and Neural Networks had high accuracy rates in grading the students based on the historical performance, engagement, and demographics.

#### **3.2 Predictive Modeling Approaches**

The predictive model of Mathematics Education Technology (MET) was created by (He, et al., 2025), and it was found that eXtreme Gradient Boosting (XGBoost) with L2 Regularization and SMOTER would yield  $R^2 = 0.842$ .



*Figure 1: Research framework of educational prediction model (He et al., 2025)*

The study by Alsbou and Alsaraireh (2024) offers an integrated approach to the process of data gathering, preprocess, data analysis, and visualization but reveals 90 percent accuracy and 87 percent sensitivity in detecting at-risk students. Their architecture is based on conventional cloud

computing with real-time prediction times of less than 2 seconds, and can be deployed by resource-constrained institutions.

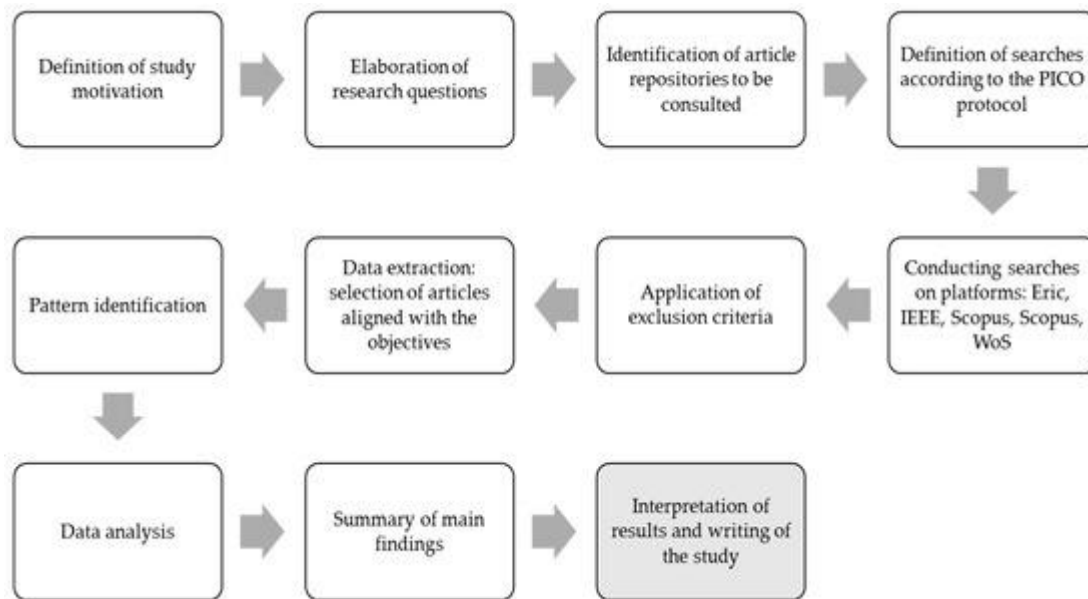


Figure 2: Main stages of research methodology

### 3.3 Deep Learning and Advanced Techniques

According to (Ahmad, et al., 2024), the usage of deep learning techniques, especially neural networks, in more complicated tasks, such as sentiment analysis, automated scoring of essays, and recognition of learning patterns, is on the rise. These are unstructured data (text, pictures, video) models that allow performing an in-depth analysis of the student behaviors. (Salas-Pilco & Xiao, 2022) emphasized that the neural network-based tutoring system is capable of creating personalized recommendations, as high as 0.3 seconds per interaction, which is significantly faster than human response time, by predicting the emotions, efficacy, and progression of students.

### 3.4 Active Learning and Optimization

This practice is implemented in the process of operating and maintaining a system and is used to develop and implement strategies to optimize the services delivered by the system to the organization.

Educational AI systems have been enhanced with advanced methods of optimization to fulfill problems of imbalance in a dataset and improvement of the performance of the model. Active Learning when combined with SMOTER and L2 regularization yielded the optimal  $R^2$  values of 0.822, compared with the 0.240 (base). The hyperparameter tuning using Particle Swarm Optimization had  $R^2 = 0.842$ , and MAE = 0.084, which is 44 percent lower than base models.

These optimization techniques are also important in practice in the classroom. (Haider & Muneer, 2025) indicated that optimization methods facilitated the correct engagement forecasts in Pakistani K-12 STEM classrooms, which yielded quantifiable changes especially with teacher training, which is an essential implementation aspect that has been disregarded in technical assessments.

#### 4. Performance Metrics and Validation

According to Ahmad et al. (2024),  $R^2$ , MAE, MSE, and RMSE are the key regression metrics; accuracy, precision, recall, and F1 scores are the key classification task metrics. He et al. (2025) highlighted that minimum of MAE in the educational contexts would take precedence because the same performance among the students cannot be ensured even under the same conditions.

**Table 5:** *AI Integration on STEM Subject Performance (2021–2025) along with 2026 Forecast*

| Year | Math<br>Performance (%) | Science Performance (%) | Physics Performance (%) |
|------|-------------------------|-------------------------|-------------------------|
| 2021 | 58                      | 61                      | 54                      |
| 2022 | 60                      | 63                      | 57                      |
| 2023 | 65                      | 70                      | 63                      |
| 2024 | 72                      | 78                      | 72                      |
| 2025 | 80                      | 83                      | 75                      |
| 2026 | 88 (Forecasted)         | 88 (Forecasted)         | 81.3 (Forecasted)       |

*Figure 3: Student performance progression in STEM subjects*

| <i>Readiness, AI Interaction, Student Engagement, and Performance</i> |                                              |                |                                                                  |
|-----------------------------------------------------------------------|----------------------------------------------|----------------|------------------------------------------------------------------|
| <b>Relationship</b>                                                   | <b>Path Coefficient (<math>\beta</math>)</b> | <b>p-value</b> | <b>Interpretation</b>                                            |
| Teacher Readiness → AI Interaction                                    | 0.45                                         | < 0.01         | Teacher readiness significantly increases AI tool interaction.   |
| AI Interaction → Student Engagement                                   | 0.38                                         | < 0.01         | AI tool interaction leads to increased student engagement.       |
| Student Engagement → Student Performance                              | 0.30                                         | < 0.05         | Increased student engagement significantly improves performance. |

Figure 4: progression of AI/LA tool integration and teacher readiness

Reliability was important in the external validation. Controlled experiments by He et al. (2025) revealed that model-directed MET obtained Hedges g effect sizes of 0.727-0.764, which was higher than the meta-analytic average of 0.677. Nonetheless, there are also methodological weaknesses: the sample size (n=45 per group) is too small to generalize, and the validation (China) is not cross-cultural, and long-term retention effects cannot be determined because of the short-term assessment (one semester). They claimed 90 percent accuracy in a variety of datasets but they

admitted that validation was done using past data and not on future deployment where drift could be experienced by the models.

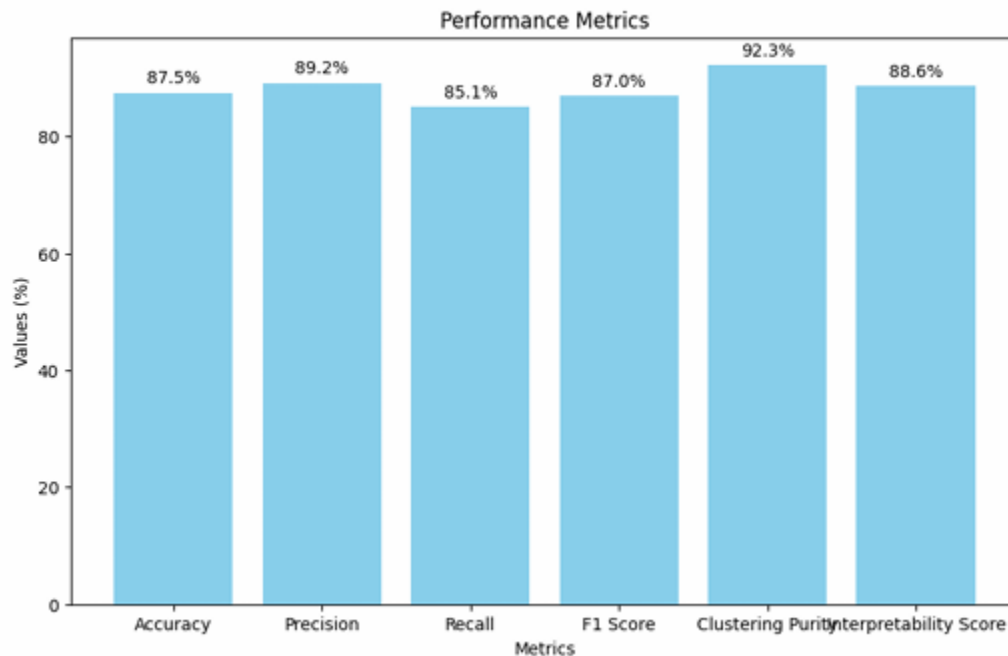


Figure 5: Performance evaluation metrics of the AI-driven framework

## 5. Analysis of Findings

The review of all the seven studies indicates that AI and learning analytics can greatly improve educational results in the case of their effective implementation. The quantitative evidence of (He, et al., 2025) has strong meta-analytic evidence (effect size  $g=0.677$ ,  $p<0.001$ ) and longitudinal findings of (Haider & Muneer, 2025) show improvements in five years. The effectiveness, depending on the context of implementation, will have a significant difference, where primary school students will have the highest effect ( $g=0.766$ ), middle school ( $g=0.695$ ), and high school ( $g=0.583$ ) students. The studies reviewed come to the point of the necessity of integrating the types of technologies with the effective pedagogical practices. He et al. revealed that AR/VR technologies had the largest effect sizes ( $g=0.788$ ), whereas constructivist teaching produced the most consistent ones ( $g=0.705$ ). The new contribution exposes that the time spent using MET has

inverted U-shaped relationships with effectiveness, and SHAP analysis shows that the most critical attribute of usage is usage time (48.5% importance). Inadequate exposure as well as overexposure diminished the benefits whereas moderate exposure (18-23 periods) maximized the results. The infrastructure, teacher training, and policy support are important implementation factors that were identified among all of the reviewed studies. According to Haider et al., the infrastructure quality (0.22,  $p < 0.01$ ) and teacher readiness (0.18,  $p < 0.05$ ) had significant predictive performance. The predictive accuracy depends on the area of application, with He et al. obtaining  $R^2 = 0.842$  whilst commercial dropout prediction models attain 82-88% accuracy. But false positive rates (8-15) require man power. The interpretability.

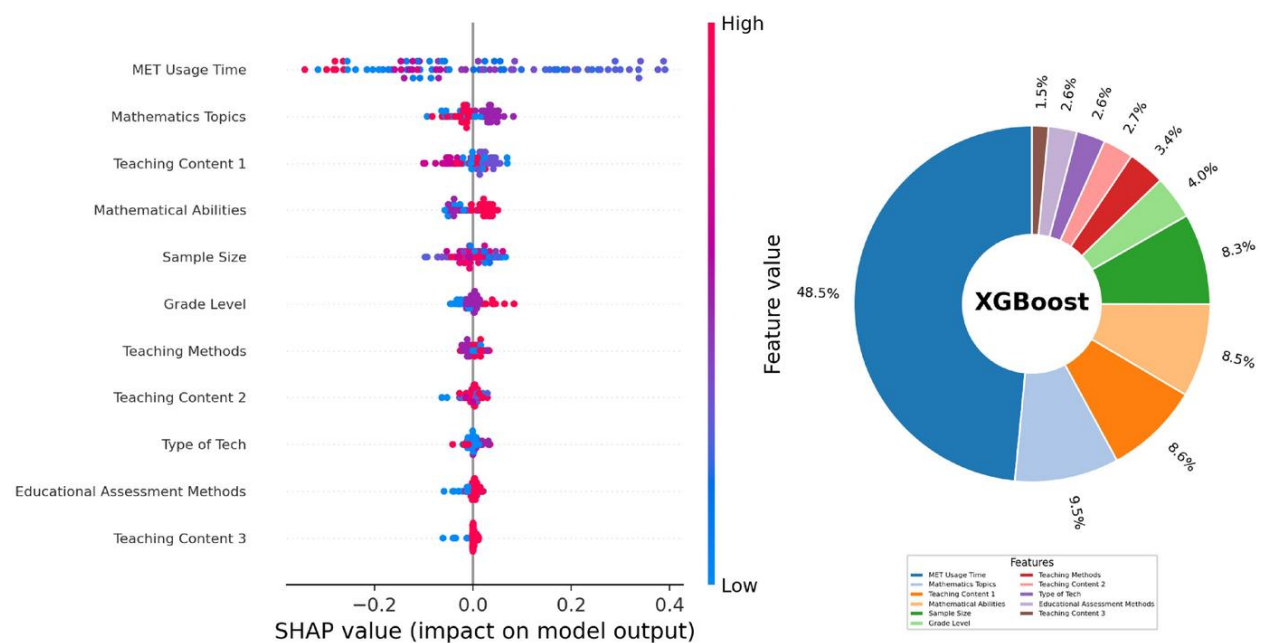
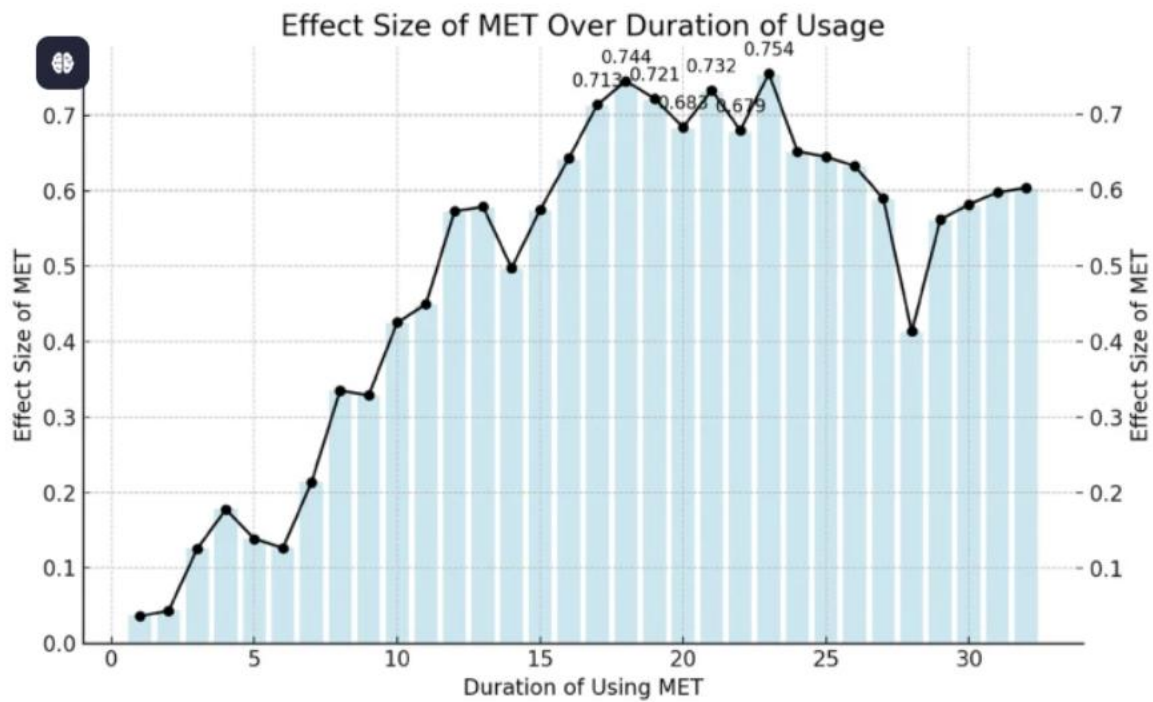


Figure 6: HAP summary plot showing feature importance

**Fig. 8**



Results of feature selection by PSO-optimized XGB with SMOTER, L2 regularization and AL

*Figure 7: Effect size of MET over duration of usage showing optimal range*

## 5. Applications of AI in Education

### 5.1 Student Performance Prediction

According to Ahmad et al. (2024), student grading and assessment were the most widely studied applications, where machine learning models could examine previous performance, attendance records, submitted assignments and participation in forums to forecast final grades. His XGBoost model, which predicts the learning effect sizes in mathematics, had  $R^2 = 0.842$  and (Alsbou & Alsaraireh, 2024) obtained 87.5% accuracy in predicting the overall performance of students in a variety of subjects.

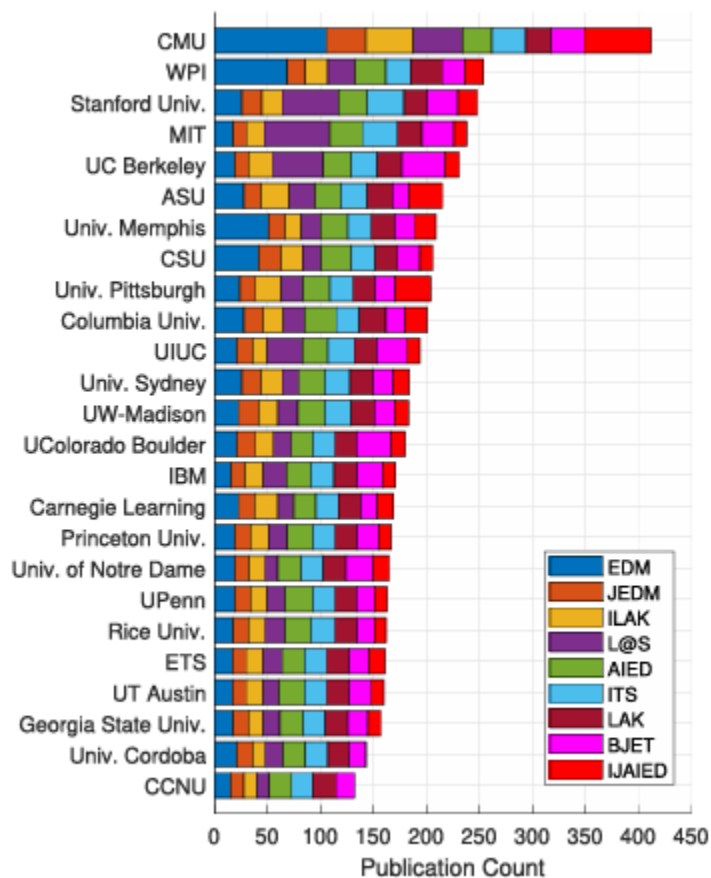


Figure 8: Institution rankings in AI education research

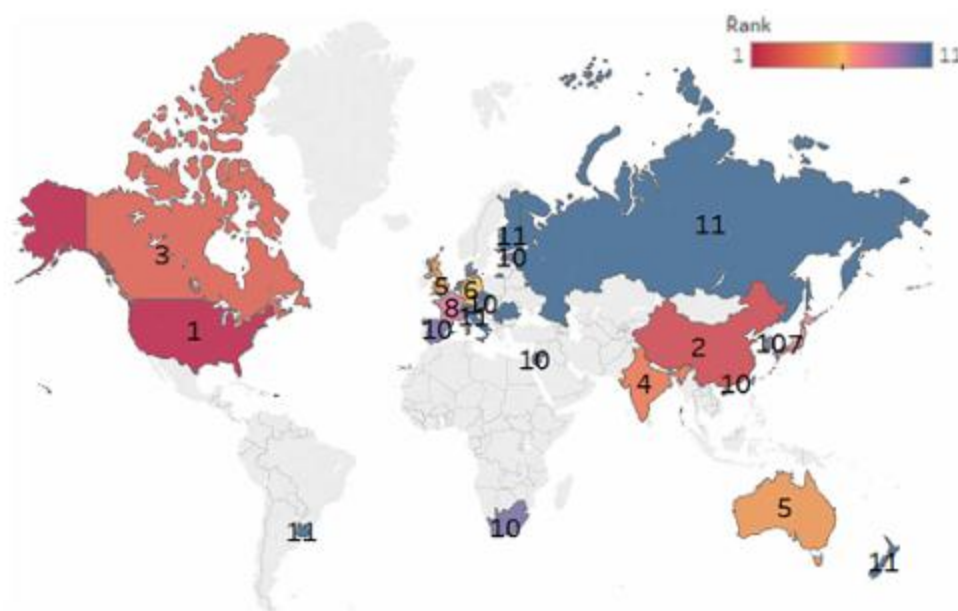


Fig. 4. Rank of countries based on their publication count in our dataset during

Figure 9: Country contributions to AI in education research

## 5.2 Personalized Learning Systems

(Salas-Pilco & Xiao, 2022) discovered that AI-based intelligent tutoring systems can solve the problem of teacher shortage in the world. Real world applications have been made in Third Space Learning (UK) and Squirrel AI (China) as well as Enlearn (USA), which serve more than 100,000 students each year. According to (Ahmad, et al., 2024) these systems use content-based, collaborative filtering, and hybrid-based systems, which are measured by grade and engagement rates of students. Economic viability EDA Cost-benefit analysis shows per-student expenses of between \$15-50/year versus \$200-500/year of human tutoring, showing it is economical at scale.

## 5.3 Learning Analytics and Data Mining

(Alsoub & Alsaraireh, 2024) have created frameworks which include machine learning to predict analytics.

(Salas-Pilco & Xiao, 2022) distinguished behavioral (access patterns, social interactions), and discourse data as the main ones. Gold-standard Analysis has the highest rate of identification (85-92%), but with a false positive rate (8-15%), the analyses require a human touch prior to interventions, which Wang (2021) is opposed to, stating that excessive dependence on algorithmic recommendations without understanding of the context may result in inappropriate interventions.

## 5.4 Decision Support Systems and Commercial Tools

(Haider & Muneer, 2025) showed that AI systems are used to select the most suitable duration of technology and they use 18 teaching periods to be the best fit in analytic geometry ( $p < 0.001$ ). Ahmad et al. (2024) conducted a review of commercial applications: assessment workload is graded in a commercial application (Grade scope, Pearson WriteToLearn automate 60-80 percent of every assessment), predictors of retention (Campus Labs, RNL Predictor, Othot achieve 82-88 percent dropout prediction accuracy), and classroom monitoring (Jibble, SAFR, Airclass real-time attendance and engagement tracking). Implementation prices are different: institutional licenses cost between 5000-50000 in the first year based on the number of students, which is 0.1-0.5 percent of the average school budgets on technology.

## 6. Challenges and Limitations

### 6.1 Infrastructure and Resource Constraints

Pakistani schools with 1:10 student to computer ratios were reported by (Haider & Muneer, 2025) at the start of the process, which increased to 1:2 in 5 years with an average of 150,000 dollars per school.

(Als bou & Alsaraireh, 2024) found that in addition to the hardware, inappropriate technical support and the lack of sustainable funding schemes impede the implementation. Resource estimation models indicate the following minimum requirements: 1 device in 4 students, 10 Mbps internet, 200/teacher spending in a year on training, and dedicated IT support (1 tech to 500 students).

## 6.2 Teacher Training and Digital Competence

(Salas-Pilco & Xiao, 2022) identified only 60 percent of the research that involved teachers and digital competence and perceptions were significant factors of adoption. (Haider & Muneer, 2025) revealed readiness gain (35% to 90% in five years) by using 40 hours of yearly training, peer mentoring and administrative assistance. (Wang, 2021) underscores that data literacy training should go beyond technical skills to involve knowledge on the limitations of AI, awareness of biases, and the ability to make pedagogical judgments when AI advice is in opposition to professional knowledge.

**Table 7.** Participants across the reviewed studies.

| Participants                      | Reference Number                                        | Number of Studies |
|-----------------------------------|---------------------------------------------------------|-------------------|
| Pre-service teachers (PSTs)       | [31,32,34,36,37,38,43,44,47,48,49,50,52,54,55,56,58,60] | 18                |
| In-service teachers (ISTs)        | [33,39,40,41,45,51,53,57,59]                            | 9                 |
| Both pre- and in-service teachers | [35,42,46]                                              | 3                 |

*Figure 10: Summary of studies on AI and learning analytics in teacher education*

## 6.3 Data Quality and Availability

(Ahmad, et al., 2024) found an extensive lack of databases to be a significant impediment. The datasets that are available (PISA, Portuguese Educational Statistics) are not in extensive contexts. (He, et al., 2025) resolved scarcity by conducting meta-analysis of 423 studies (38,163 students), which is associated with the problem of publication bias and heterogeneity. Reliability was proved via their bias analysis (funnel plots, Fail-Safe N=69,413, trim-and-fill tests), but the issues of temporal validity have been identified due to the rapid development of educational settings.

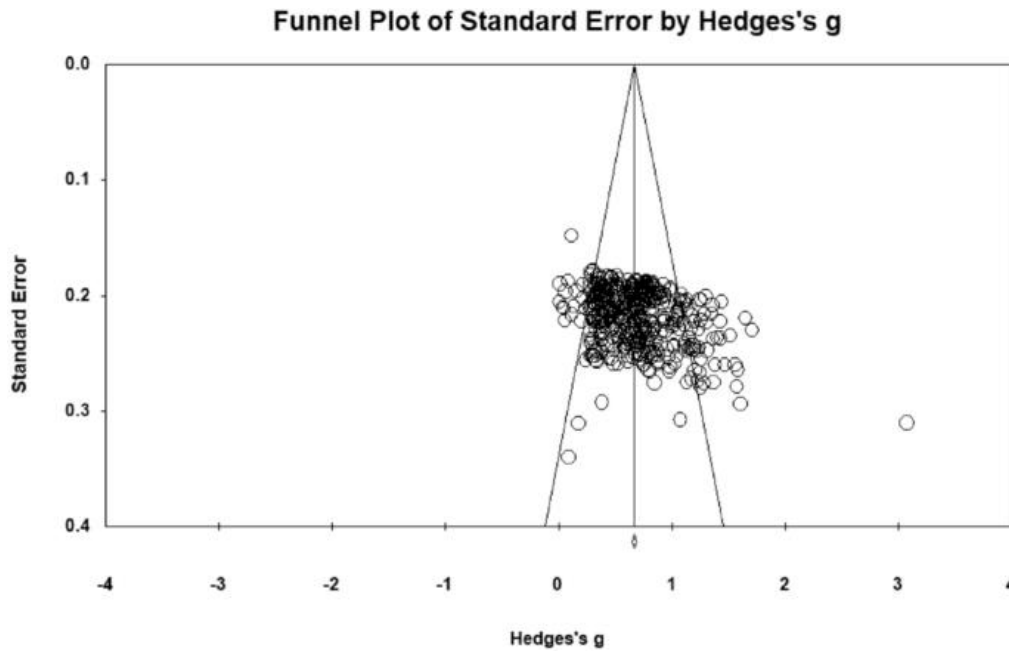


Figure 11: Funnel plot for publication bias assessment

#### 6.4 Model Interpretability and Trust

According to (Ahmad, et al., 2024), the traditional forecasting based on the experience of educators was hard to accept by the stakeholders in the high-stakes decision-making, and the black box nature of AI is quite difficult. He et al. (2025) used SHAP values that indicate that MET usage time showed the first place of importance (48.5%), mathematics topics (9.5%), and teaching content (8.6%). (Wang, 2021) states that technical transparency is not sufficient but teachers need pedagogically meaningful interpretations that are in line with the realities in the classroom, saying that the informed are prepared to show appreciation of algorithms, even when they are incorrect, due to the existence of algorithms.

#### 6.5 Amplified Biases and Discrimination

(Wang, 2021) offers a highly important piece of evidence that AI can strengthen the existing bias in the educational decision-making. The use of gender bias in hiring algorithms was trained on male-dominated data sets, and facial recognition algorithms proved to be erroneous in regard to race and gender. The AI risk-assessment algorithms were discriminatory towards minority groups. Wang cautions that biased training data may effectively place students of minorities, low

socioeconomic families and students with special needs in vicious cycles of marginalization instead of equity.

## **7. Datasets and Methodologies**

They analyzed the literature that used different datasets to train AI models and validate them. (He, et al., 2025) used PRISMA-based systematic search of the CNKI, Web of science, Springer, Science Direct, and Taylor and Francis databases and gathered information on 423 studies.

PRISMA 2020 flow diagram for updated systematic reviews which included searches of databases, registers and other sources

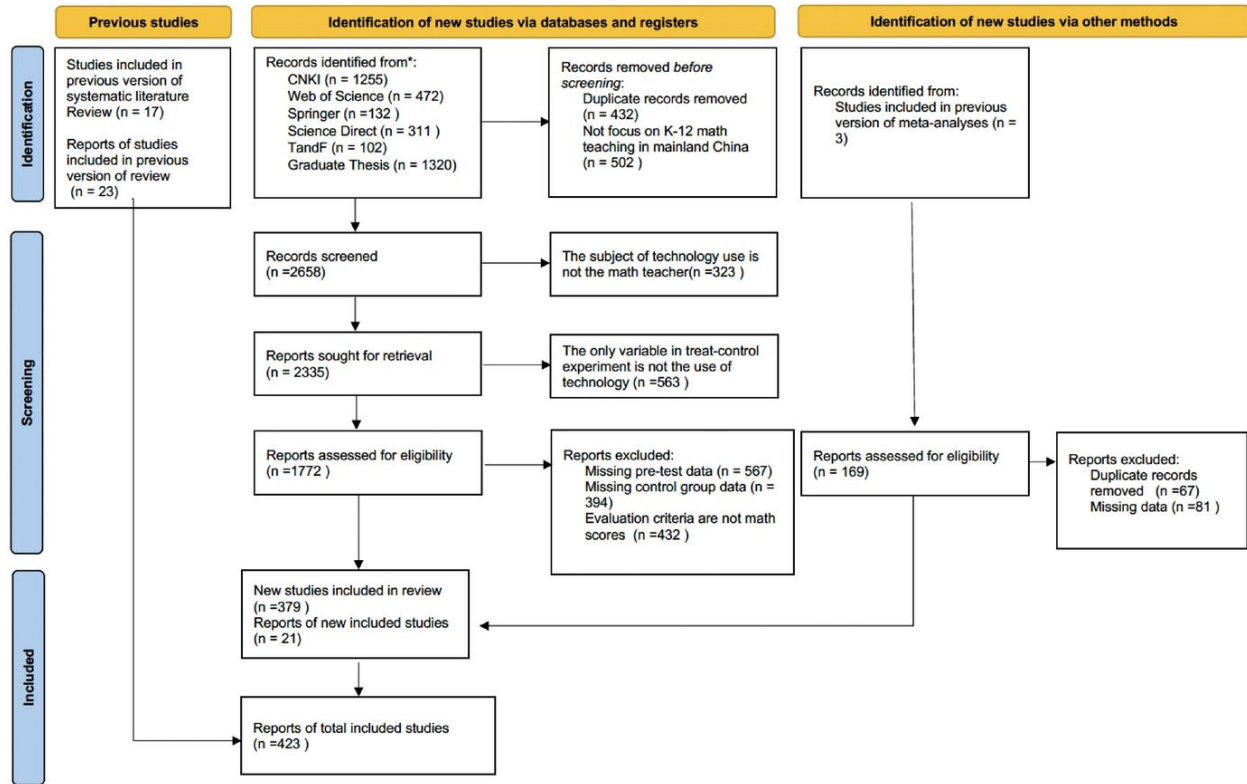


Figure 12: PRISMA flow diagram for literature search and selection

(de Souza Zanirato Maia, et al., 2023) searched databases of ERIC, IEEE, Scopus, Science Direct, and Web of Science. (Ahmad, et al., 2024) performed a bibliometric review on 4,447 articles on 9 major journals published between 2014-2022. There were also common publicly available datasets of PISA (Programme for International Student Assessment) of OECD and Portuguese Educational Statistics.

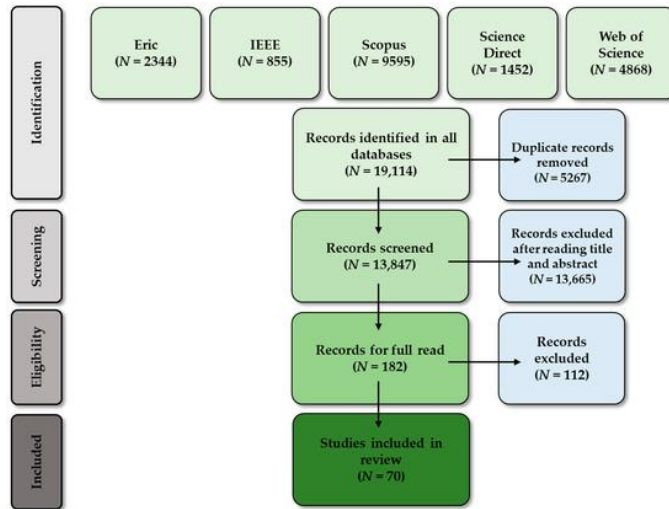


Figure 13: PRISMA flow diagram for systematic review

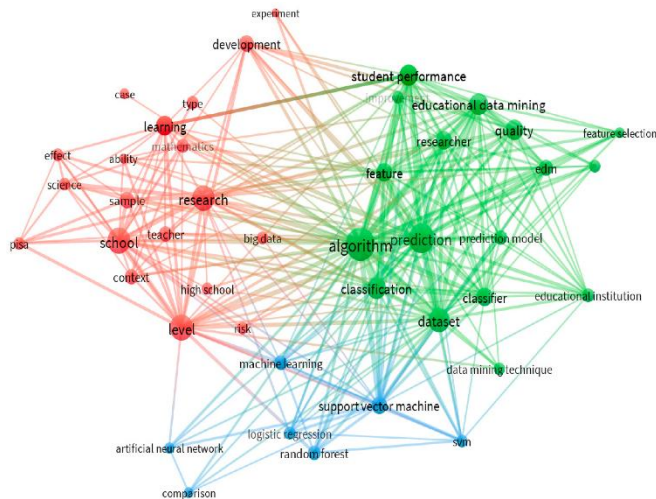


Figure 14: Network map of research terms

(Haider & Muneer, 2025) used longitudinal data to study 6 schools during a period of five years, whereas (Alsbou & Alsaraireh, 2024) created a synthetic dataset to test the construction of frameworks. (Salas-Pilco & Xiao, 2022) observed that behavioral data of the learning management system, discourse data of the online forums, and statistical data (demographics and performance indicators) were used in most studies.

## 8. Future Research Directions

### 8.1 Emerging Technologies

(Ahmad, et al., 2024)proposed generative AI (GPT-4, Claude) as disruptive and provides an array of general-purpose functions by prompting and fine-tuning instead of creating specific models. Initial applications in automated feedback, lesson planning support and adaptive content generation are promising. (He, et al., 2025)proposed to combine them: reinforcement learning, deep learning to tutor, NLP and computer vision to fully analyze the classroom.

### 8.2 Personalization and Adaptive Learning

The development of personalization features is currently one of the essential research concerns of educational AI. (Ahmad, et al., 2024)highlighted the issue that the existing systems suggest the content and make it harder or easier, but creating really individualized materials in real-time is complex. The areas where the optimization of learning schedule based on AI modelling considering the principles of cognitive psychology, the complexity of the topic, stages of student development, and the availability of resources should be a focus of future research.

### 8.3 Explainable AI Development

One of the most important research directions in the creation of trustworthy and easily adoptable AI systems is the development of more interpretable and transparent systems. The authors of SHAP value showed that the utility can demonstrate feature importance (MET usage time 48.5%). (Salas-Pilco & Xiao, 2022)emphasized that educators need not just expectations but a description of why an intervention is suggested and how it can be applied to the specific environment. Enhancing the need of public scrutiny, (Wang, 2021) argues that in the absence of transparency, AI will be a consolidation of power. Pedagogical knowledge should be combined with predictive abilities and be integrated in future systems to offer actionable and contextual information.

### 8.4 Educational Equity and Cultural Responsiveness

Educational Equity and Cultural Responsiveness Educational equity and cultural responsiveness can be described as equitable education provision in all aspects of diversity and culturally sensitive education provisions.

The role of future studies should be to focus on the educational equity issues and make AI benefits accessible to the underserved population. The systematic integration demonstrated by (Haider & Muneer, 2025) is useful in resource-constrained settings but underlines the need to persist with the policy support, financial investments, and capacity development to avoid the further division. According to (Ahmad, et al., 2024), a majority of meta-analyses do not consider geographic and cultural factors. The impacts of a digital divide are also recorded to be massive by (Wang, 2021) and access inequalities have a disproportionate impact on the lower social and economic settings. Future studies need to create culturally inclined AI that is sensitive to both linguistic and occasional practices and educational demands.

## 9. Ethics, Legal and Social Considerations

### 9.1 Privacy and Regulatory Compliance

(Salas-Pilco & Xiao, 2022)disclosed that 60 percent of the studies did not perform any ethical procedures, and this data protection issue was serious. (Wang, 2021)cautions against the possibility of re-identification of anonymized data, i.e., in a so-called anonymized data, one can be singled out as a vulnerable population, such as undocumented immigrants and religious minorities, can be singled out unfairly.

### 9.2 Algorithmic Bias and Fairness

According to the authors, discriminatory training data have the potential to undermine underrepresented groups (Ahmad, et al., 2024). (Alsbou & Alsaraireh, 2024)included the elements of fairness measurements and bias control. (Wang, 2021)stresses that educational leaders have to be aware of their bias blind spots and work with stakeholders as a group.

### 9.3 Moral Decision-Making and Human Oversight

(Wang, 2021)advances such a serious contention: the utilitarian decision-making that AI is used in contradicts the moral grounds of education. AI does not possess empathy that is needed to make decisions that are compassionate. Closures of schools due to data alone are a representation of cold utilitarianism communities that individuals view as unfair. Wang claims that AI can only be used as a decision support and not a replacement, and the leaders should remain responsible to making compassionate judgment.

### 9.4 Transparency and Accountability

(He, et al., 2025)have shown transparency in SHAP, however, the technical transparency is not enough without the interpretations that have pedagogical meaning. According to (Ahmad, et al., 2024)the clear accountability frameworks should stipulate the developer, institutional, and educator roles in the monitoring systems, responding to the issue, and making high-stakes decisions, which should be grounded in human judgment. (Wang, 2021)stresses that authority

becomes more and more represented in an algorithmic form, but most of them are still proprietary black box algorithms that cannot be analyzed, and it is challenging to evaluate the truthfulness, determine fairness, or reach equity especially when the algorithms reinforce educational inequality.

## 9. Conclusion

This review of seven studies indicates that AI and learning analytics can be highly useful in educational decision-making under the condition that they are adopted properly, and meta-analytical results indicate that the effect sizes can be substantial ( $g=0.677$ ,  $p<0.001$ ) and longitudinal research reveals that improvements can be maintained throughout STEM fields. The synthesis determines the best implementation strategies such as proper technology-pedagogy interactions and AR/VR and constructivist techniques appear to have the most significant results and indicates important timing factors in which moderate technology use (18-23 periods) is better than the use of minimal or excess exposure.

Nevertheless, to be able to successfully integrate AI, it is necessary to deal with significant obstacles such as infrastructure limits, teacher training requirements, data quality threats, and ethical issues related to privacy and algorithm bias. Experience in resource-constrained environments shows that with systematic professional development and long-term policy funding, the implementation issues can be addressed, but geographical inequalities remain with the underrepresentation of developing areas in the research of AI education.

In the future, explainable AI development, learning equity, and culturally responsive systems should be prioritized and leave high-stakes choices to human oversight. The analyzed sources all lead to the same conclusion whereby technology needs to be combined with pedagogical skills, moral accountability, and humanitarian judgments. In a world where AI is taking control of educational decision-making, teachers should be cautious of the potential of artificial intelligences to boost or undermine the decision-making processes, be it by fostering increased bias or dehumanization, and should use AI as a means to augment and not to eliminate the human element in assisting students.

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